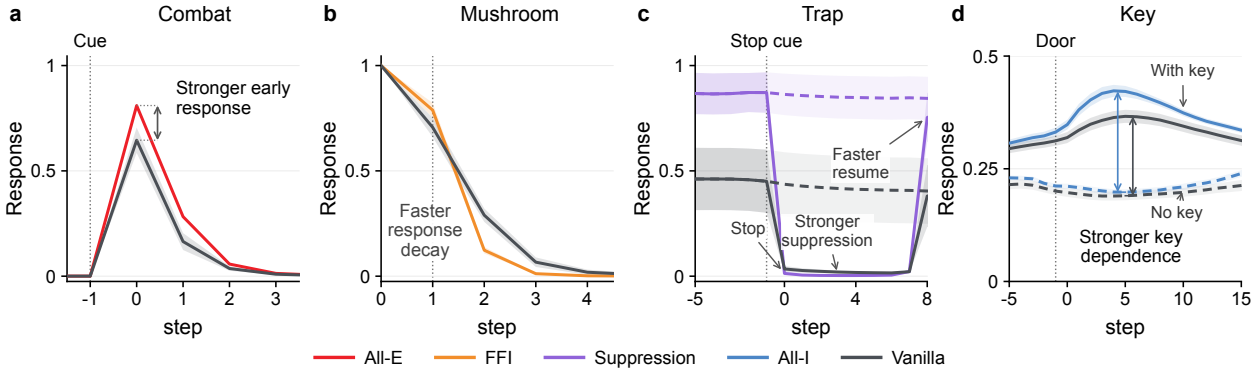



**Figure R3 | Task-relevant response dynamics.**

a-d, Combat, Mushroom, Trap and Key: FFL models and Vanilla. Response denotes the tendency to advance in Combat, approach a mushroom in Mushroom, continue movement in Trap, or pass through a door in Key, measured by the probability assigned to the corresponding actions. Mushroom responses are normalized to onset. Solid/dashed curves: Go-Stop/Go-only in c; key-present/key-absent in d. Lines and shading: mean ± 1 SEM across 10 training seeds. a, The shortened time window highlights the early peak; the bracket marks the All-E minus Vanilla mean response at step 0. b, The time window spans steps 0-4.5 to show the initial decline and residual response on one axis. Faster response decay refers to the initial decline in approach probability after the poisonous mushroom cue. c, Stronger suppression refers to lower stopped responses relative to each model's Go-only condition. d, The response axis spans 0-0.5. Coloured arrows show the key-present minus key-absent mean response at step 5 (All-I, 0.223; Vanilla, 0.176); arrows are offset horizontally for clarity, with guides to the same sampled step.